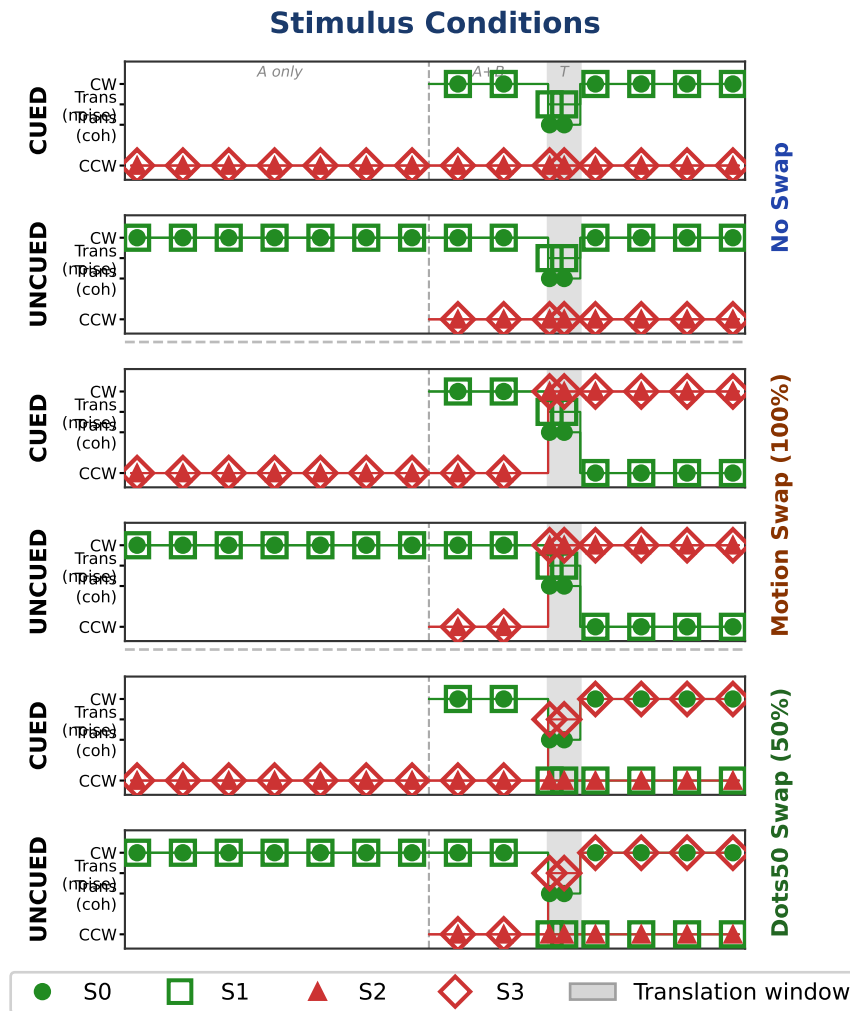


# VRDots — Flat Swap Conditions: Motion Swap & Dots50 Swap

Sessions: 260324\_1010 (Motion Swap) 260325\_1039 (Dots50 Swap)

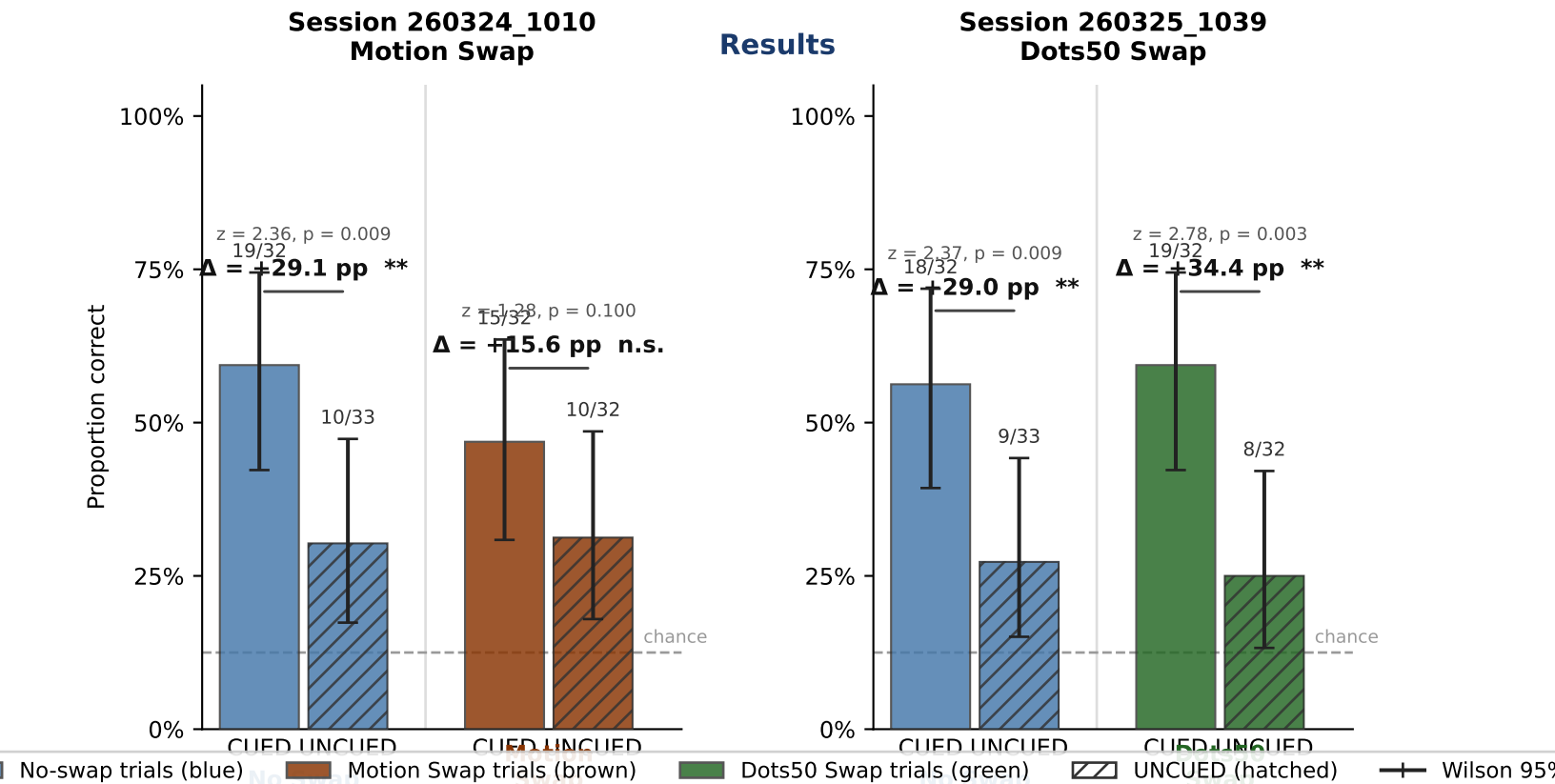
Two sessions examined how within-trial manipulations applied at translation onset affect the cueing effect. Both used a flat (no depth), 750 ms delayed-onset design with 80 ms translation and 8-AFC direction judgment at 2.0 m. Each session comprised 128 trials interleaving swap and no-swap trials in equal proportion (64 each), fully crossing Cond (CUED/UNCUED) × DelayedFieldColor (G/R) × RotConfig (0/1) × TransDeg (8 directions), yielding 8 trials per cell per swap type. Session 260324\_1010 tested the Motion Swap: at translation onset the non-translating field immediately reversed its rotation direction, and the translating field landed in the opposite rotation post-translation (both fields exchanged rotation identities). Session 260325\_1039 tested the Dots50 Swap: the noise subfield of each rotation group (S1 and S3) exchanged group membership at translation onset, so 50% of each field's dots changed affiliation while the coherent subfields (S0 and S2) remained unaffected.

Trajectory diagrams below show one representative color-rotation pairing ( $S0\bullet/S1\square$  = initially CW field;  $S2\blacktriangle/S3\blacklozenge$  = initially CCW field; green = CW, red = CCW). Both color assignments and both rotation configurations were balanced across trials. Dashed vertical line = delayed field onset (~750 ms); shaded band = 80 ms translation window.



## VRDots — Flat Swap Conditions (continued)

Analysis: proportion correct was computed for each Cond × SwapType cell, pooled across color, rotation, and translation direction ( $n = 32$  trials per cell, except one N/UNCUED cell with  $n = 33$ ). The cueing effect is the difference in accuracy (CUED – UNCUED, in percentage points). A one-tailed z-test assessed whether CUED > UNCUED within each swap type. A second z-test compared cueing effects between swap and no-swap conditions (two-tailed, assessing attenuation).



### Findings

Session 260324\_1010 — Motion Swap: the cueing effect was present under both swap types. No-swap trials replicated the baseline: CUED > UNCUED by approximately 27 pp. Motion-swap trials showed a reduced but still positive cueing effect of approximately 16 pp. The ~11 pp attenuation is consistent with the idea that continuity of the pre-translation rotation history contributes to the onset cue: when the non-translating field adopts the translating field's rotation identity at onset, the distinctiveness of the cue is reduced. However, cueing was not eliminated, indicating that the temporal onset asymmetry alone (independent of sustained rotation identity) provides residual attentional benefit.

Session 260325\_1039 — Dots50 Swap: the 50% dot-membership swap produced no measurable attenuation of cueing. The cueing effect was approximately 30 pp without swap and approximately 34 pp with Dots50 swap — a slight non-significant increase. The Dots50 swap selectively reassigns the noise subfields (S1, S3) while leaving the coherent subfields (S0, S2) intact. The null result suggests that partial disruption of dot grouping at the noise level does not impair attentional selection, and that the remaining 50% coherent-subfield continuity is sufficient to maintain a perceptually unified surface.

### Notes on data quality

Notes on data quality: (1) Session 260324\_1010 was collected the same morning as the 260324\_0716 baseline in which the cursor-jump artifact was first identified (thumbstick snapping to adjacent 45° sectors, producing ~25% adjacent-sector errors). Whether the hysteresis fix (33° latch, two-stage confirmation) was active for this session is uncertain; if the artifact persisted it would have reduced accuracy uniformly and attenuated the cueing effect without directional bias. Session 260325\_1039 was almost certainly collected after the fix. (2) At  $n = 32$  trials per cell, Wilson 95% CIs are approximately  $\pm 18$  pp, so fine-grained comparisons (e.g., the ~11 pp motion-swap attenuation) are suggestive but not decisive at this power level. Both results should be treated as directional signals pending replication with larger  $n$ .